TRIBHUVAN UNIVERSITY

INSTITUTE OF SCIENCE AND TECHNOLOGY

HIMALAYA COLLEGE OF ENGINEERING

CHYASAL,LALITPUR

Lab Report No:- 4

Title:- IMPLEMENTATION OF Additive & Multiplicative inverse in Z_n and testing coprimality using Euclidean Algorithms.

Submitted By:-

Name :- Anis Pal

Roll No:- 5

Submitted To:-

Department Of CSIT

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LAB4: IMPLEMENTATION OF Additive & Multiplicative inverse in Zn and Testing Coprimality using Euclidean Algorithms.

Objectives:

1. To compute the additive inverse of a number in Zn.
2. To calculate the multiplicative inverse of a number in Zn using the extended Euclidean Algorithm.
3. To determine whether two positive integers are relatively prime using the euclidean Algorithm.

Theory:

1. Additive inverse in Zn

In modular arithmetic, the additive inverse of a number a in Zn is a number b such that:

$$a + b \equiv 0 \pmod{n}$$

The additive inverse can be easily computed as:

$$b = n - a \text{ (if } a \neq 0)$$

2. Multiplicative inverse in Zn

The multiplicative inverse of a in Zn is a number x such that:

$$a \cdot x \equiv 1 \pmod{n}$$

A multiplicative inverse exists only if a & n are coprime (i.e, $gcd(a, n) = 1$).

The extended Euclidean Algorithm is used to find integers x & y satisfying.

$$ax + ny = gcd(a, n)$$

if $gcd(a, n) = 1$, then x is the multiplicative inverse modulo n.

3.) Relatively prime (coprime) Numbers

Two integers a & b are relatively prime if their greates common divisor (gcd) is 1.

The euclidean Algorithm efficiently computes gcd(a,b) using repeated division.

$$gcd(a,b) = gcd(b, a \bmod b)$$

if gcd(a,b) = 1, then the numbers are coprime.

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Q.1.) Addative inverse of number in Zn.

```
#include <stdio.h>
 int n, a, i, j, additive_inverse[100], inverse;
 printf("Enter modulo value n:");
 scanf("%d",&n);

for(i=0;i<n;i++)
{ for(j=0;j<n;j++)
  {inverse=(i+j)%n);
   if (inverse==0)
     additive_inverse[i]=j;
   }
 }
printf("Enter the number to find additive inverse:");
scanf("%d",&a);

Printf("The additive inverse of %d is %d.", a, additive_inverse[a]);
Printf("\n Name: Anil pal  Roll no: 5");
return 0;
}
```

```
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Enter modulo value n:20
Enter the number to find additive inverse:9
The additive inverse of 9 is 11.
Name:Anil Pal ROll no:5
Process returned 0 (0x0)   execution time : 7.633 s
Press any key to continue.
```

.2.) Multiplicative inverse of any number in Zn using extended Euclidean algorithm.

```
#include<stdio.h>
int main()
{ int r1, n, b, r2, t1=0, t2=1, q, r, t, inverse;
  printf("~~t2b4~~ Enter modulus n : ");
  scanf("%d", &n);
 printf("Enter number to find multiplicative inverse:");
 scanf("%d", &b);
  r1=n;
  r2=b;
 printf("\n step \t r1 \t r2 \t q \t r \t t1 \t t2 \t t \n");
  int step=1;
  while (r2>0)
  { q=r1/r2;
    r=r1-q*r2;
     t=t1-q*t2;
    printf("%d \t %d \t %d \t %d \t %d \t %d \t %d \t %d \n");
     r1=r2;
     r2=r;

     t1=t2;
     t2=t;
     step++;
     }
```

```
if (r1 != 1)
{
printf("\n Multiplicative inverse does not exist.\n");
}
else
{inverse = b1 % n;
 if (inverse < 0)
    inverse = inverse + n;
 printf("\n The multiplicative inverse of %d is %d\n", b, inverse);
}
printf("\nName: Anil Pal Roll: 5");
return 0;
}
```

```
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Enter modulus n: 21
Enter number to find multiplicative inverse: 10
The multiplicative inverse of 10 is 19.
Name:Anil Pal Roll:5
Process returned 0 (0x0)   execution time : 4.006 s
Press any key to continue.
```

5) Two positive integer are relatively coprime or not using euclidean algorithm.

```
#include<stdio.h>
int main()
{ int n1, n2, gcd, r1, r2, q, r;
 printf("enter two numbers to find if its relatively prime number:");
scanf("%d %d", &n1, &n2);
if (n1>n2){
    r1=n1;
    r2=n2;
    }
else {
  r1=n2;
  r2=n1;
  }
while (r2>0){
  q = r1/r2;
  r= r1-q*r2;
  r1=r2;
  r2=r;
 }
 gcd=r1;
if (gcd==1){
   printf("The GCD of %d and %d is %d.So, they are relatively
          prime number", n1, n2, gcd); }

else {
   printf("The GCD of %d & %d is %d .so they are not relatively
          prime number", n1,n2,gcd);

   }printf("\n Name:Anil Pal Roll:5");
    return 0;
   }
```

```
Enter two numbers to find if its relatively prime number:13 29
The GCD of 13 and 29 is 1. So, they are relatively prime number.
Name:Anil Pal Roll:5
Process returned 0 (0x0)   execution time : 5.287 s
Press any key to continue.
```

DISCUSSION

In this experiment, three related problems Additive, multiplicative inverse & coprime check are discussed & implemented in C Programming. The program calculates the additive inverse of a given number a in Zn using the simple formula n-a. And, the program uses the extended Euclidean Algorithm to find the multiplicative inverse of a modulo n. if gcd(a,n) = 1, the multiplicative inverse exists and is printed. & also we check two numbers are relatively prime or not.

CONCLUSION

In the last of lab, we successfully implemented:

1. Calculation of additive inverse in Zn.
2. Determination of multiplicative inverse using Extended Euclidean algorithm.
3. checking whether two numbers are relatively prime using Euclidean algorithm.

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